

TRAFFIC ANALYSIS

COMP6348001
NETWORK FORENSICS

SUBJECT MATTER EXPERT

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Learning Objectives

By the end of this session, you will be able to:

- Explain what traffic analysis is and why it is important in forensics.
- Differentiate between normal and abnormal network traffic.
- Identify patterns that reveal scanning, tunneling, or exfiltration.
- Apply traffic analysis concepts to real-world forensic cases.

What is Traffic Analysis?

- Examining **communication patterns** in network data, not just content.
- Focus on **who talks to who, how often, how much, and when.**
- Useful even when **encryption hides payloads.**
- Goal: detect anomalies that may signal attacks or suspicious activity.

Sources for Traffic Analysis

- **Packet Captures (pcap):** detailed, but heavy.
- **Flow Records (NetFlow, sFlow, Argus):** summaries of sessions.
- **Firewall and IDS Logs:** show allowed/blocked connections.
- **Statistical Data:** packet/byte counts, timing intervals.

Normal vs Abnormal Traffic

- **Normal:** predictable patterns, business applications, known domains, balanced request/response.
- **Abnormal:** sudden spikes, one-way flows, repeated failed attempts, traffic at unusual hours, connections to unknown IPs.

Key Features in Traffic Analysis

- **Volume:** large uploads/downloads may mean exfiltration.
- **Timing:** bursts at odd hours suggest automation.
- **Direction:** inbound vs outbound, lateral movement inside LAN.
- **Frequency:** repeated short connections can mean scanning.
- **Duration:** long sessions may indicate tunneling or persistent connections.

Attack Signatures in Traffic

- **DoS/DDoS:** massive floods of traffic, often from many sources.
- **Scanning:** sequential probes of IP ranges and ports.
- **Data Exfiltration:** steady outbound flow to external IPs.
- **Tunneling:** ICMP/DNS traffic with abnormal length or frequency.
- **Beaconing:** regular small packets to C2 server.

Example: Port Scanning

- **Pattern:** many short connections to different ports on same IP.
- **Forensic Value:** reveals reconnaissance before exploitation.
- **Detection:** abnormal frequency, SYN packets without full handshake.

Example: Data Exfiltration

- **Pattern:** steady outbound transfer, often at off-hours.
- **Forensic Value:** confirms sensitive data leaving the network.
- **Detection:** abnormal upload volumes, connections to rare destinations.

Example: DNS Tunneling

- **Pattern:** high volume of DNS queries with long/encoded subdomains.
- **Forensic Value:** attackers hiding data inside DNS requests.
- **Detection:** abnormal length, frequency, entropy of DNS queries.

Case Study: Indonesian FinTech Breach (Fictionalized)

- Attackers used **DNS tunneling** to leak customer PII.
- **Evidence Sources:** DNS logs + NetFlow showed abnormal queries at 2 AM.
- **Result:** 15 GB exfiltrated before detection.
- **Lesson:** Even when encrypted, traffic patterns expose hidden channels.

Case Study: International Example – Target Breach (2013)

- Attackers compromised vendor accounts, installed malware.
- **Evidence Sources:** NetFlow + IDS logs showed C2 traffic beaconing every 90 minutes.
- **Result:** 40M credit cards stolen.
- **Lesson:** Small, repetitive connections often mean beaconing C2 traffic.

Tools for Traffic Analysis

- **Wireshark:** packet-level traffic analysis.
- **nfdump / Argus:** flow-level summaries.
- **Bro/Zeek:** advanced traffic inspection.
- **SIEM platforms:** correlation and anomaly alerts.

Best Practices

- Always **baseline normal traffic** first.
- Correlate multiple sources (packets + flows + logs).
- Look for **outliers in timing, volume, frequency, direction.**
- Document findings for repeatability.

Discussion Case

- **Scenario:** You find 300MB of outbound traffic from an HR server to an IP in Eastern Europe at 3 AM. Firewall logs say “allowed,” NetFlow shows session lasted 20 minutes.

Questions for students:

- Is this normal? Why/why not?
- What further evidence would you collect?

Sample Answer:

- Suspicious because HR servers don’t normally send bulk data externally.
- Collect DNS queries, packet captures, and check if the IP is blacklisted.

Reflection Question

When payloads are encrypted (HTTPS, VPN), do you think traffic analysis alone is enough to detect attacks? Why or why not?

Sample Answer:

- Not always enough to see content, but patterns (volume, timing, direction, frequency) often reveal malicious behavior.
- Should be combined with endpoint and application logs for confirmation.

Hands-on

1. Open b6300a.cap
2. How many UDP conversations are in the trace file?
 - a. 33
 - b. 37
 - c. 44
 - d. 22
3. What is the most active host?
 - a. 172.31.19.73
 - b. 172.31.19.54
 - c. 224.0.1.35
 - d. 172.31.19.255
4. Were all SNMP requests successful?
5. How many requests were rejected?
 - a. 2
 - b. 4
 - c. 3
 - d. 5
6. How many bytes do the ICMP packets have as a payload?
 - a. 28
 - b. 30
 - c. 24
 - d. 8
7. Why is the option to follow the UDP stream available when clicking on the ICMP packet?

Hands-on

1. Open telnet-raw.pcap
2. Look at the first packet. What is the real sequence number?
 - a. 72603759
 - b. 0
 - c. 20385233
 - d. 91048592
3. Why did the sequence number increase in packet number 3?
4. In the trace file, there are some TCP keep-alives. Why are they generated?
5. Does this connection support SACK?
6. Is Window scaling supported? What is the scaling value?
7. Keep-alive packets have the PSH bit set to 1. Why is this so?
8. Why did the sequence number increase?

Homework

1. Start capturing your own traffic
2. Open a command line, and start a ping to www.google.com
3. Initiate a traceroute to www.google.com
4. Stop the capture
5. Look at the traceroute traffic. How is the TTL field changing?
6. What payload do your ICMP packets generate?
7. Why can't you follow the stream to see the payloads?

SEE YOU NEXT WEEK!